UNIVERSITY OF TECHNOLOGY AND APPLIED SCIENCES-NIZWA

INFORMATION TECHNOLOGY DEPARTMENT

Sample Final Examination
CSCN1104-Computer Network Fundamentals

MAX. MARKS **40** **TIME ALLOCATION** **2 Hour**

STUDENT ID	ANSWER-KEY	STUDENT NAME	ANSWER-KEY
SECTION		LEVEL	Diploma First Year

INSTRUCTIONS TO THE STUDENTS:

- Understand the questions properly before answering.
- Time Management is very important. Write to-the-point answers. Do not write unnecessary details.
- The Question Paper consists of **SEVEN (7)** Pages including the cover page.
- Use the backside of the page for calculations and extended answers.
- Calculators are allowed.

Question Type	Total Marks	Marks Obtained
Multiple Choice Questions	10.00	
True or False	7.00	
Fill in the blanks	4.00	
Match the following	5.00	
Short Answer Questions	6.00	
Scenario Based Question	8.00	
TOTAL MARKS	40.0	

Academic Integrity:

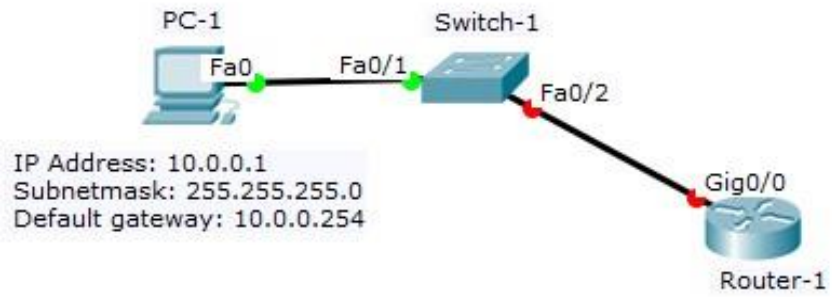
Any form of cheating should be avoided. Anyone caught violating will be penalized according to the college by-laws and academic integrity policy.

Checked By	Rechecked By	Verified By

SECTION-A: MULTIPLE CHOICE QUESTIONS (10 X 1 Mark = 10.0 Marks)

Select the best answer

#	Questions
A1	**Which of the following command is used to set the device name on a router as CCLab1 ?** a. Devicename CCLab1 b. Host Name Router CCLab1 c. Hostname CCLab1 d. Name CCLab1
A2	**Which one of the following methods is used to securely establish a remote CLI connection through a virtual interface, over a network?** a. Console b. Telnet c. AUX d. SSH
A3	**The command "*ip default-gateway* "is used to configure _____.** a. Default-gateway on the router b. Default-gateway on the switch c. Default-gateway on the PC d. Default-gateway on the IP phone
A4	**Which of the following type of information is included in the transport layer header?** a. Destination and source logical address b. Destination and source port numbers c. Destination and source physical address d. Destination and source PC numbers
A5	**Which one of the following is the binary equivalent of decimal number 195?** a. 01011111 b. 00011110 c. 11111100 d. 11000011
A6	**If one bit is borrowed from the host portion of a class-C address, what would be the number of usable host addresses?** a. 128 b. 126 c. 254 d. 2

A7	**Which of the following IPv6 migration techniques allows the existence of both the IPv4 and IPv6 addresses on the same network?** a. Tunnelling b. Dual Stack c. Translation d. Encryption
A8	 **From the above exhibit, the address 10.0.0.254 should be configured on _____.** a. Fa0/1 interface of Switch-1 b. Fa0/2 interface of Switch-1 c. Gig0/0 interface of Router-1 d. None of the above
A9	**Which of the following command is used to test the connectivity between two hosts?** a. nslookup b. Ping c. Netstat d. ipconfig
A10	**Which of the following portions of an IPv6 address is assigned by the service provider to the customer?** a. Interface ID b. Global Routing Prefix c. Subnet ID d. Local ID

SECTION-B: TRUE OR FALSE (7 X 1 Mark = 7 Marks)

#	Questions	Answer
B1	**The default gateway address is not required, if both the source and destination hosts are on the same network.**	**True**
B2	**The number of usable host addresses on a network that has a /24 mask are 256.**	**False**
B3	**The role of a router is to select the best path and direct packets towards the destination host.**	**True**
B4	**A network prefix value /27 indicates that there are 27 bits for host and 5 bits for network.**	**False**
B5	**192.168.200.25 is a private class-C IPv4 address.**	**True**
B6	**The Application, Presentation and Session layer of OSI model is equivalent to the Application layer of TCP/IP model.**	**True**
B7	**SMTP is an example of web protocol.**	**False**

SECTION-C: FILL IN THE BLANKS (4 X 1 Mark = 4 Marks)

#	Questions
C1	**The result of ANDing 1 and 0 is _____0_____.**
C2	**The network prefix value for the subnetmask 255.255.255.252 is _____/30_____.**
C3	**The command that displays the contents of running configuration file on a router is _____Show running-config_____**
C4	**_____IPv6_____Version of IP addresses are in the size of 128 bits.**

Section – D Match the Following (5 x 1 = 5 Marks)

Match the Column – A with Column – B.

#	Column-A	#	Column-B
D1	**Enable**	A	**Interface configuration mode**
D2	**Exit**	B	**Used to move directly from Line configuration mode to privilege exec mode**
D3	**Configure terminal**	C	**Used to move from user exec mode to privilege exec mode.**
D4	**Specific configuration mode for configuring IP address on a switch.**	D	**Used to move from Global configuration mode to privilege exec mode**
D5	**End**	E	**Used to move from privilege exec mode to Global configuration mode**

QUESTION	D1	D2	D3	D4	D5
ANSWER	C	D	E	A	B

SECTION-E: SHORT ANSWER QUESTIONS (6.0 Marks)

1. **Briefly differentiate between Local host and Remote host.**

Local host: This is a host on the same local network as the sending host. The hosts share the same network address. Local hosts can reach each other and share information without the need for any additional devices. If a host is sending a packet to a device that is configured with the same IP network as the host device, the packet is simply forwarded out of the host interface, through the intermediate device, and to the destination device directly.

Remote host: This is a host on a remote network. The hosts do not share the same network address. Devices that are beyond the local network segment are known as remote hosts. When a source device sends a packet to a remote destination device, then the help of routers and routing is needed.

2. **Subnet the address 192.168.15.0 to provide 6 subnets, with each subnet containing 25 usable host addresses, and fill up the given table.**

Total Number of Subnets	**8**
Total number of usable host addresses	**30**
Custom Subnet Mask	**255.255.255.224**
Number of bits borrowed	**3**

3. **Compress the following IPv6 address 2001:0DB8:0000:0000:0000:1111:0000:0200 by applying the two rules.**

Rule1: 2001:DB8:0:0:0:1111:0:200

Rule2: 2001:DB8::1111:0:200

Section – F
Scenario Based Question (8 - Marks)

1. **You are setting up a medium sized network for an organization with the class-C IP address 192.168.99.0/24. The organization has two departments, Sales and Finance respectively. The Sales department requires 20 usable host addresses and Finance department requires 6 usable host addresses. Divide the network address 192.168.99.0/24 to provide the required addresses to ensure minimum wastage of addresses.**

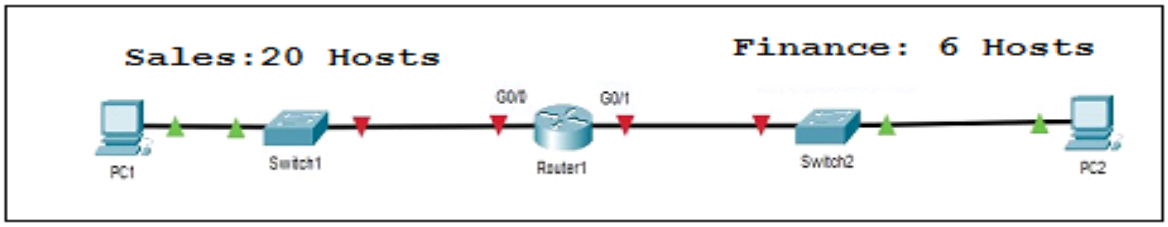
   Read the following instructions and fill up the required details in the boxes of subnet table given below

   - **Assign the first available subnet to the Sales department.**
   - **Assign the next available subnet to the Finance department.**

Subnet Table:

Department Name	Network Address	First Host Address	Second Host Address	Last Host Address	Broadcast Address	Subnet Mask
Sales	192.168.99.0	192.168.99.1	192.168.99.2	192.168.99.30	192.168.99.31	255.255.255.224
Finance	192.168.99.32	192.168.99.33	192.168.99.34	192.168.99.38	192.168.99.39	255.255.255.240

Using the above subnet table, complete the addressing table by referring the topology and instructions given below:



Instructions:

Assign the first host address from Sales subnet to G0/0 interface of Router1.

Assign the last host address from Sales subnet to PC1.

Assign the second host address from Sales subnet to VLAN 1 interface of Switch1.

Assign the first host address from Finance subnet to G0/1 interface of Router1.

Assign the last host address from Finance subnet to PC2.

Assign the second host address from Finance subnet to VLAN 1 interface of Switch2.

Addressing Table:

Device	Interface	IP Address	Subnet Mask	Default Gateway
Router1	G0/0	192.168.99.1	255.255.255.224	NA
	G0/1	192.168.99.33	255.255.255.240	NA
Switch1	VLAN 1	192.168.99.2	255.255.255.224	192.168.99.1
Switch2	VLAN 1	192.168.20.34	255.255.255.240	192.168.99.33
PC1	NIC	192.168.99.30	255.255.255.224	192.168.99.1
PC2	NIC	192.168.99.38	255.255.255.240	192.168.99.33